A COMPARATIVE STUDY OF MACHINE LEARNING METHODS FOR URL-BASED PHISHING DETECTION

7030 CEM CYBER SECURITY INDIVIDUAL PROJECT

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INTRODUCTION

- Website Phishing is a cyber security fraud in which the attacker will be faking a legitimate website by its content.
- Also known as URL Phishing
- In 2022, Anti- Phishing Working Group (APWG) observed 1,025,968 phishing attacks.
- IBM’s 2022 cost of data breach report found that the average cost of data breach increased from $4.24m in 2021 to $4.35m in 2022.
- Detected using HTML code, URL etc

OBJECTIVES

- Collect phishing and legitimate URL data using python web scraping tools and extract the URL features from the dataset.
- Clean, preprocess, prepare the dataset to train the ML models.
- Develop and train the model using prepared data.
- Test and evaluate the performance .of the machine learning models on different parameters.

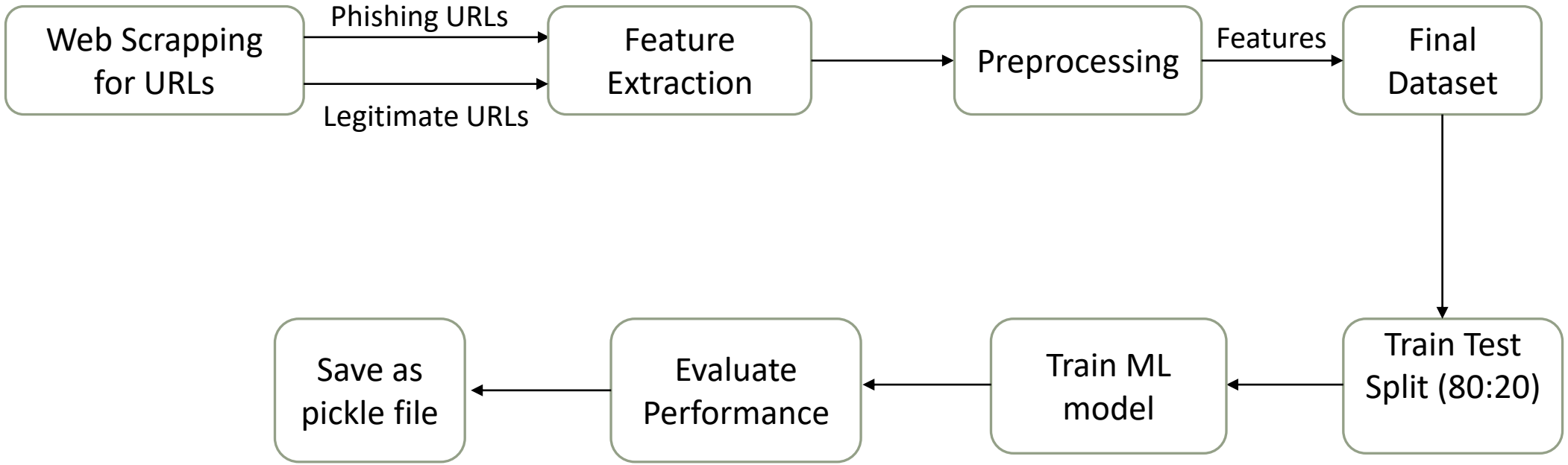
LITERATURE SURVEY

Reference	Approach Used	Number of Features	Dataset Used	Accuracy	Pros	Cons
[1]	Random Forest, Gradient Boosting, SVM (Support Vector Machine)	22	1056937	SVM - 99.896%	• High results in handling the imbalanced dataset	• Difficulty in training
[2]	Logistic Regression, K neighbors Classifier, Random Forest, Decision Tree, and XGBoost	48	Million samples	XGBoost – 99%	• High performance when more data used as training data	• Million samples - difficulty in training
[3]	Decision Tree (DT), Random Forest (RF), XGBoost, Multilayer Perceptrons, K-Nearest Neighbors, Naive Bayes, AdaBoost, and Gradient Boosting	87	11430	XGBoost - 96.4%	• Faster training and testing • Reduced overfitting	• Small dataset-URLs

Reference	Approach Used	Number of Features	Dataset Used	Accuracy	Pros	Cons
[4]	Naïve Bayes algorithm	22	5471	74%	• Modified algorithm • Less features extracted	• Old dataset • Increase the URL data training set
[5]	SVM, Naïve Bayes, Decision Tree, K-NN, Adaptive Random Forest algorithm (ARFA)	63	96012	ARFA - 96.91%	• Uses modified algorithm • Low error rate	• Difficulty in training as number of samples is more
[6]	ML and Advanced Deep Learning models		Millions of samples	DNN - 96.38%	• It offers ease of use and implementation	• Time complexity • Limited storage
[7]	Long short-term memory (LSTM) , convolutional neural network (CNN) and LSTM-CNN	80	20000	CNN – 99.2%	• Relevant only to URL characteristics of the same dataset.	• Does not check status of URL website • Low speed

Reference	Approach Used	Number of Features	Dataset Used	Accuracy	Pros	Cons
[8]	Decision tree, Support Vector Machine, Naïve base, 2D and 3D Neural Networks, TensorFlow Artificial General Intelligence (AGI)	30	30000		• Improved efficiency and efficacy • Feature elimination	• Low recognition speed
[9]	Decision Tree (DT), Random Forest (RF), Logistic Regression, Naïve Bayes algorithm	15	126077	Random Forest - 83%	• Balanced and enhanced dataset can create better solutions	• Longer execution time
[10]	Random Forest, Naïve Bayes, SVM, XGBoost, Gradient Boosting	23	13000	XGBoost - 99.8%	• High accuracy achieved in less time	• Limited set of features

ARCHITECTURE



METHODOLOGY

1. Data collection using web scraping and other sources

2. Feature extraction

3. Data cleaning and preprocessing

4. Develop ML models

5. Train ML models

6. Evaluate the performances using metrics

7. Test and evaluate the best model

DATA COLLECTION

- 2 types of URLS:
  - **Legitimate URL** - Easy resources such as google index, alexa etc.
  - Legitimate URL source - Cyber Security research of University of New Brunswick (https://www.unb.ca/cic/datasets/url-2016.html).
  - 35337 legitimate URLs collected from University of New Brunswick
  - **Phishing URL** - Phishing URLs should be live/fresh/latest
  - 78505 phishing URLs collected from Phishtank.com

FEATURE EXTRACTION

1) Address Bar Based Features –

- Domain of URL
- IP Address in URL
- “@” symbol in URL
- Length of URL
- Depth of URL
- Redirection “//” in URL
- “http / https” in Domain name
- Using URL shortening services “TinyURL”
- Prefix or suffix “-” in Domain

FEATURE EXTRACTION contn…

2) Domain Based Features –

- DNS Record
- Website Traffic
- Age of Domain
- End period of Domain
- Page Rank
- Google index
- Status Report

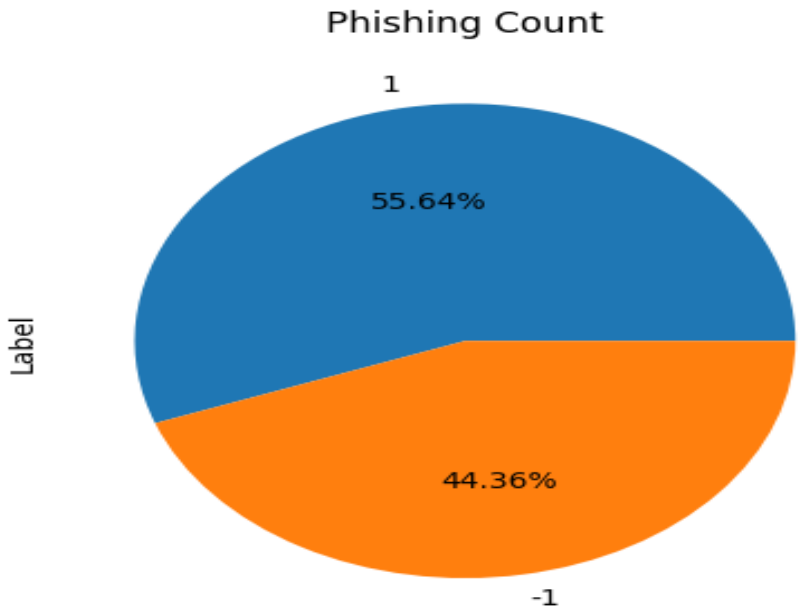
FEATURE EXTRACTION contn…

3) HTML & Javascript Related Features -

- IFrame Redirection
- Status Bar Customization
- Disabling Right Click
- Website Forwarding
- Link in script Tag
- Server from handler
- Using pop-up windows
- Link pointing to page
- HTTPS response
- Non –standard port
- Info mail
- Favicon
- Request URL
- Abnormal URL
- Anchor URL

MACHINE LEARNING MODEL BUILDING

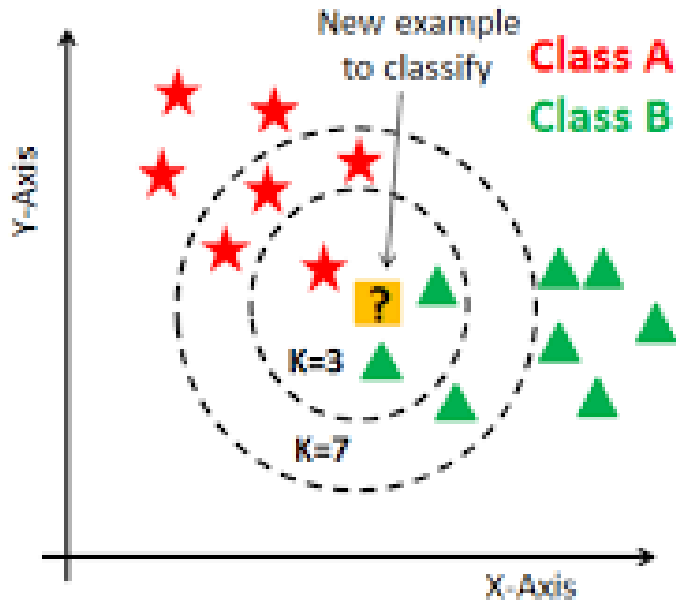
- Took 10000 data to make training easy.
- No outliers and missing values.
- Split the data into training set and test set in the ratio 80:20.

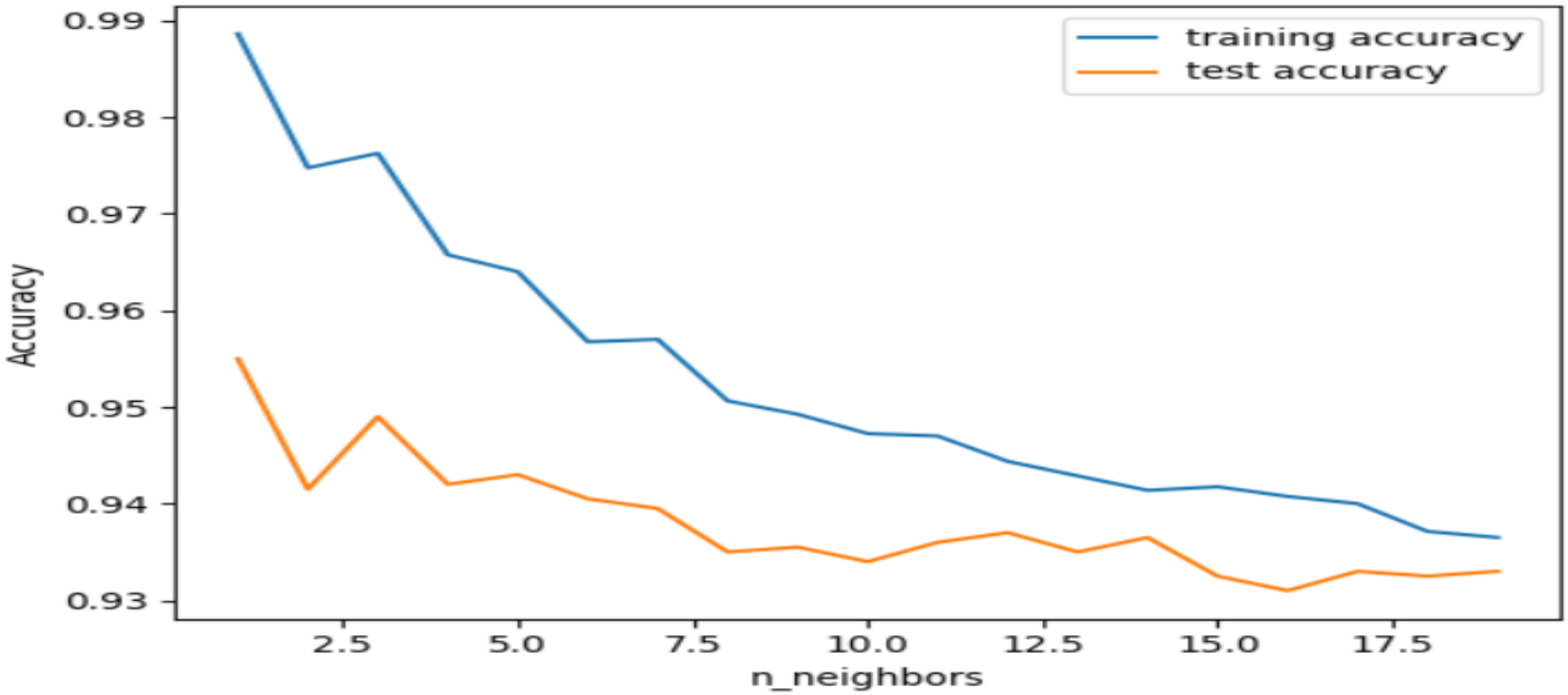


MACHINE LEARNING MODELS

1) MODEL 1 : KNN (K- NEAREST NEIGHBOR) ALGORITHM

- A supervised ML algorithm for classification.
- It makes predictions based on the similarity of data points to other neighboring data points.




training accuracy
test accuracy
0.99
0.98
0.97
0.96
0.95
0.94
0.93
Accuracy
2.5
5.0
7.5
10.0
12.5
15.0
17.5
n_neighbors

```
K-Nearest Neighbors : Accuracy on training Data: 0.989
K-Nearest Neighbors : Accuracy on test Data: 0.955

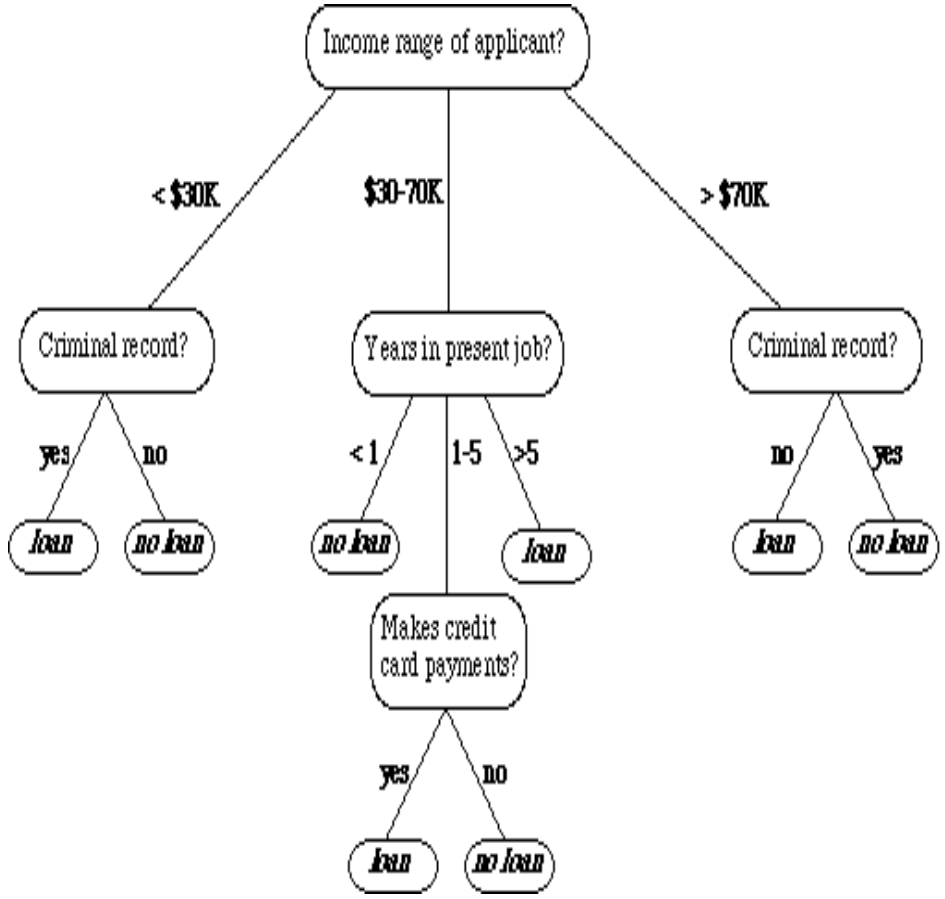
K-Nearest Neighbors : f1_score on training Data: 0.990
K-Nearest Neighbors : f1_score on test Data: 0.960

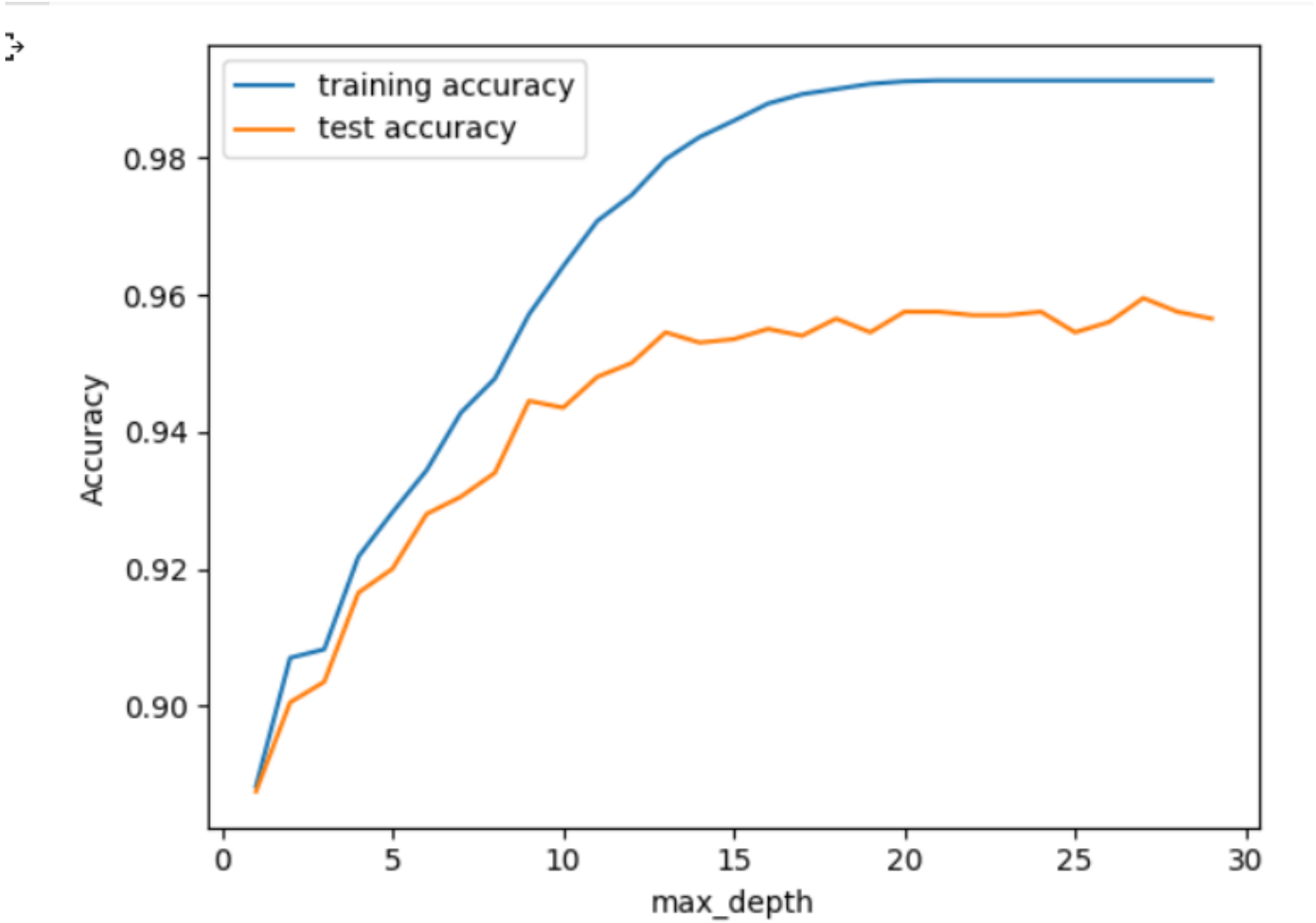
K-Nearest Neighbors : Recall on training Data: 0.990
Logistic Regression : Recall on test Data: 0.962

K-Nearest Neighbors : precision on training Data: 0.990
K-Nearest Neighbors : precision on test Data: 0.957
```

MODEL 2: DECISION TREE ALGORITHM

- A condition-based algorithm.
- Checks the condition is met or not. Then decide the output.




training accuracy
test accuracy
0.98
0.96
0.94
0.92
0.90
Accuracy
0
5
10
15
20
25
30
max_depth

```
Decision Tree : Accuracy on training Data: 0.991
Decision Tree : Accuracy on test Data: 0.958

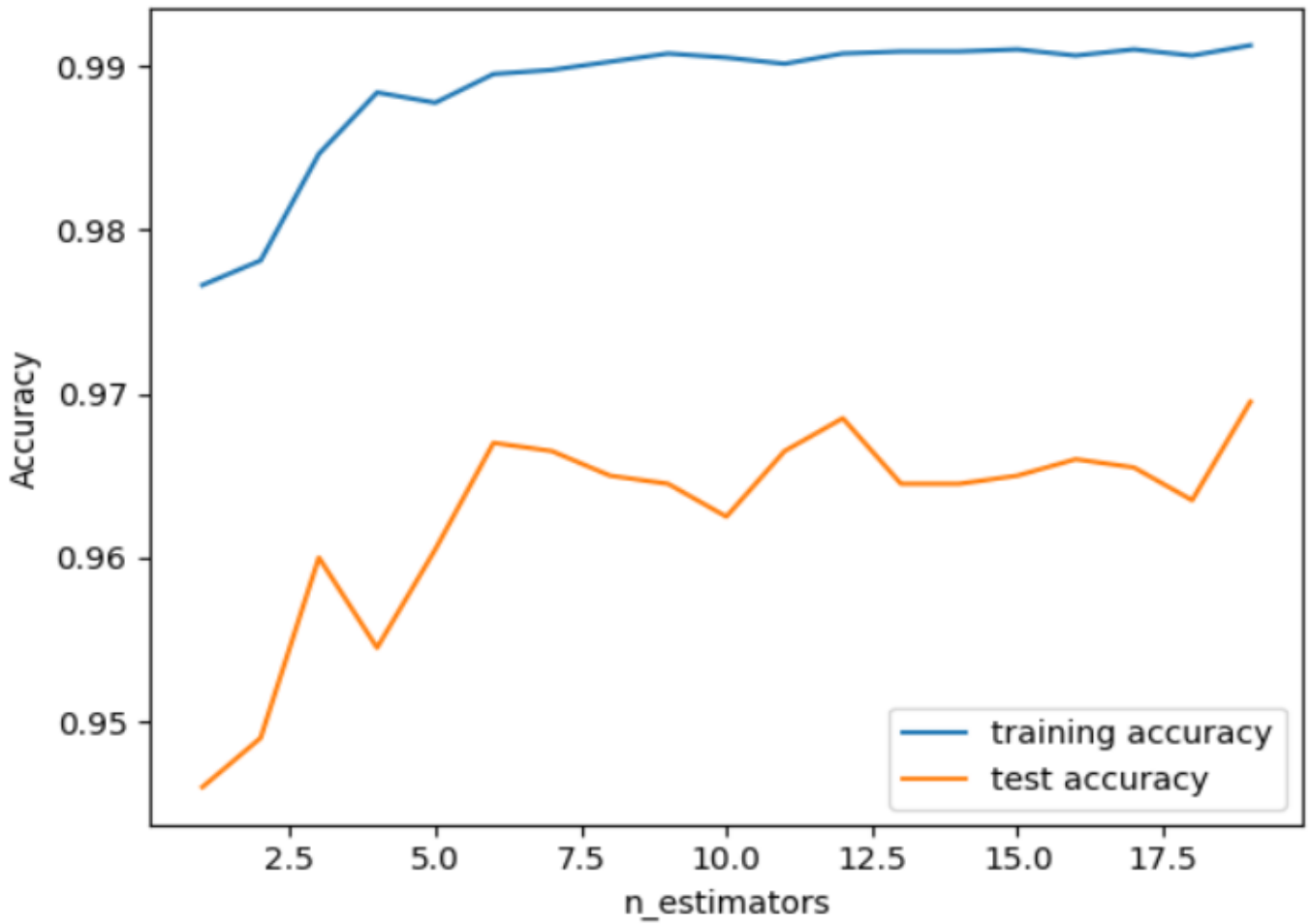
Decision Tree : f1_score on training Data: 0.992
Decision Tree : f1_score on test Data: 0.962

Decision Tree : Recall on training Data: 0.992
Decision Tree : Recall on test Data: 0.962

Decision Tree : precision on training Data: 0.993
Decision Tree : precision on test Data: 0.962
```

MODEL 3: RANDOM FOREST

- A supervised ML algorithm for classification.
- An ensemble algorithm in which a number of models make prediction at the same time.
- Random forest = n number of decision trees making predictions
- It use decision tree algorithms as the base algorithm.
- We define n = number of decision trees to make predictions and make final prediction based on voting (the class predicted by more trees).


0.99
0.98
0.97
0.96
0.95
Accuracy
2.5
5.0
7.5
10.0
12.5
15.0
17.5
n_estimators
training accuracy
test accuracy

```
Random Forest : Accuracy on training Data: 0.991
Random Forest : Accuracy on test Data: 0.969

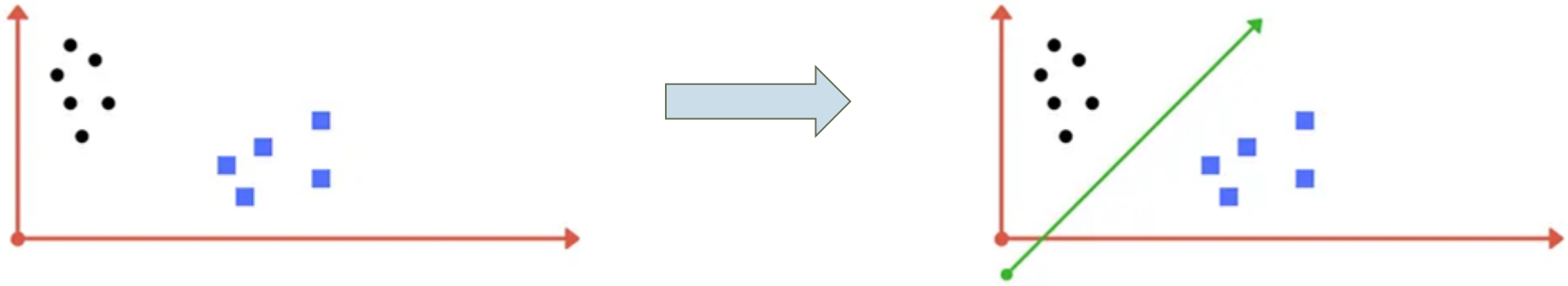
Random Forest : f1_score on training Data: 0.992
Random Forest : f1_score on test Data: 0.972

Random Forest : Recall on training Data: 0.994
Random Forest : Recall on test Data: 0.971

Random Forest : precision on training Data: 0.990
Random Forest : precision on test Data: 0.962
```

MODEL 4: SVM

- A Classic ML model that optimizes a hyperplane (support vector) that separates two classes of data.
- The algorithm finds an optimal hyperplane for the training data.
- It require very less training data to achieve good accuracy.



```
Support Vector Machine : Accuracy on training Data: 0.969
Support Vector Machine : Accuracy on test Data: 0.956

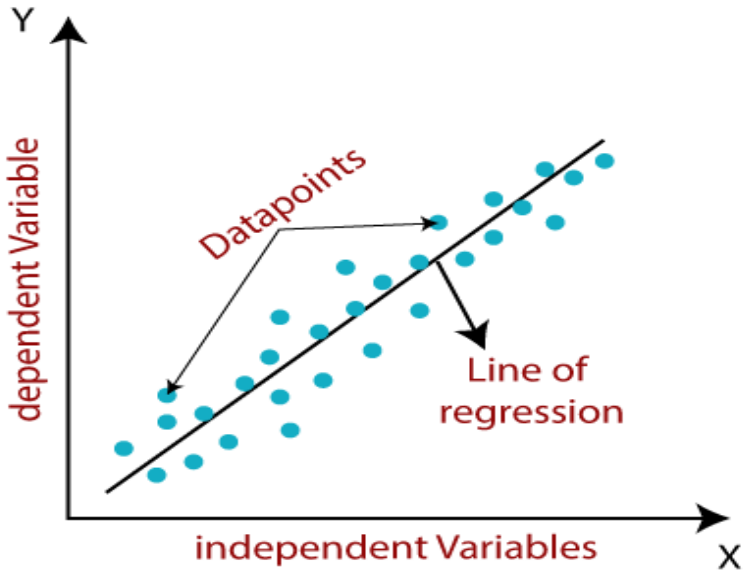
Support Vector Machine : f1_score on training Data: 0.973
Support Vector Machine : f1_score on test Data: 0.961

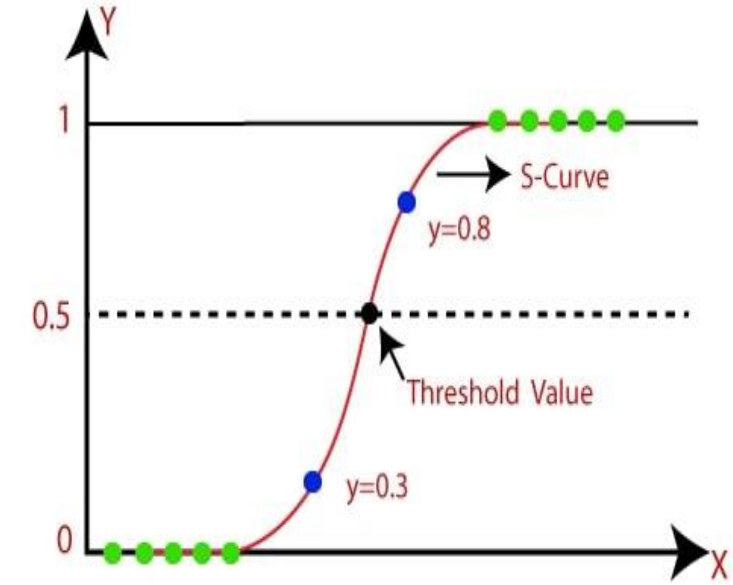
Support Vector Machine : Recall on training Data: 0.979
Support Vector Machine : Recall on test Data: 0.967

Support Vector Machine : precision on training Data: 0.966
Support Vector Machine : precision on test Data: 0.955
```

MODEL 5: LOGISTIC REGRESSION

- Linear Regression - Find a line that passes closer to all possible data points.
- Logistic Regression - Linear Regression plus a probability threshold





```
Logistic Regression : Accuracy on training Data: 0.927
Logistic Regression : Accuracy on test Data: 0.924

Logistic Regression : f1_score on training Data: 0.935
Logistic Regression : f1_score on test Data: 0.932

Logistic Regression : Recall on training Data: 0.944
Logistic Regression : Recall on test Data: 0.940

Logistic Regression : precision on training Data: 0.927
Logistic Regression : precision on test Data: 0.925
```

MODEL 6: XGBOOST



```
XGBoost Classifier : Accuracy on training Data: 0.988
XGBoost Classifier : Accuracy on test Data: 0.965

XGBoost Classifier : f1_score on training Data: 0.989
XGBoost Classifier : f1_score on test Data: 0.969

XGBoost Classifier : Recall on training Data: 0.992
XGBoost Classifier : Recall on test Data: 0.992

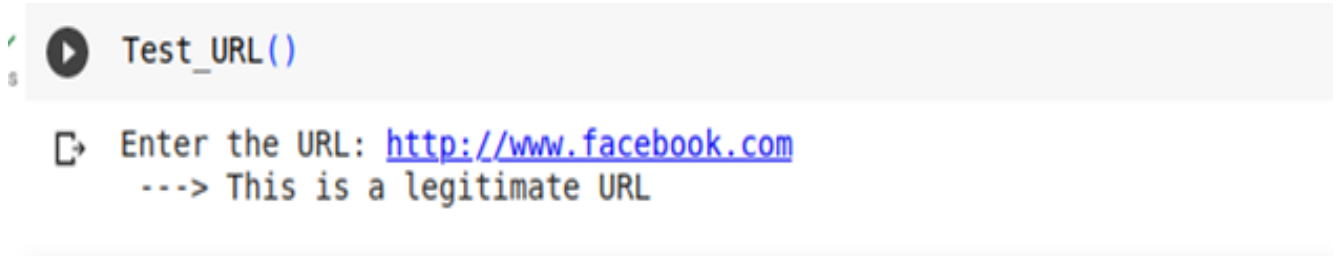
XGBoost Classifier : precision on training Data: 0.986
XGBoost Classifier : precision on test Data: 0.986
```

COMPARISON

	ML Model	Accuracy	f1_score	Recall	Precision
0	Random Forest	0.968	0.971	0.993	0.990
1	XGBoost Classifier	0.965	0.969	0.992	0.986
2	Decision Tree	0.957	0.961	0.992	0.993
3	Support Vector Machine	0.956	0.961	0.979	0.966
4	K-Nearest Neighbors	0.955	0.960	0.990	0.990
5	Logistic Regression	0.924	0.932	0.944	0.927

RESULT


Test_URL()
12s
Enter the URL: http://www.famipay.famldigi.bsjdw55z.top/
---> This is a phishing URL!


Test_URL()
Enter the URL: http://www.facebook.com
---> This is a legitimate URL

CONCLUSION

- The extracted features play important role in the detection accuracy.
- 6 ML models were trained in real time.
- Random forest has the best testing accuracy followed by XGBoost and Decision Tree.
- Logistic Regression has the lowest performance.

FUTURE SCOPE

- The Accuracy of the models can be increased by extracting more features.
- Other Deep Learning can be used for training and check if they outperform the ML model.
- The developed model can be hosted as a callable API so that anyone can access it anytime for phishing detection or can be used at the backend of a software as well.
- The trained model can be deployed as a browser extension (such as Chrome extension) so that phishing can be detected in real time.

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